

Secondary Examination, 2017

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 80

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) Candidate must write first his/ her Roll No. on the question paper compulsorily.
- 2) All the questions are compulsory.
- 3) Write the answer to each question in the given answer- book only.
- 4) For questions having more than one part the answers to those parts are to be written together in continuity.
- 5) If there is any error/difference/contradiction in Hindi and English versions of the question paper of Hindi version be treated valid.
- 6)

Section	Q. Nos.	Marks per question
A	1-10	1
B	11-15	2
C	16-25	3
D	26-30	6
- 7) There are internal choices in Q. Nos. 28 and 30.
- 8) Write on both sides of the pages of your answer-book. If any rough work is to be done, do it on last pages of the answer-book and cross with slant lines and write 'Rough Work' on them.
- 9) Draw the graph of Q. No. 26 on the graph paper.

Part –A

Question.1 HCF and LCM of two integers are 12 and 336 respectively. If one integer is 48, then find another integer.

Solution.

Simplify the expression,

We know that

$$LCM \times HCF = 1(\text{integer}) \times 2(\text{integer})$$

$$HCF = 12,$$

$$LCM = 336,$$

$$1(\text{integer}) = 48$$

$$2(\text{integer}) = ?$$

$$LCM \times HCF = 1(\text{integer}) \times 2(\text{integer})$$

$$= 336 \times 12 = 48 \times x$$

$$x = \frac{336 \times 12}{48} = 84$$

$$x = 84$$

Hence, second integer is 84.

Question.2 Find the sum of the first 20 terms of the A.P 13,8,3.....

Solution.

Simplify the expression,

We know that the sum of nth terms of AP.

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

The given value AP of 13, 8, 3

$$S_n = 20$$

$$n = 20$$

$$d = 8 - 13 = -5$$

$$a = 13$$

We using formula sum of nth term of AP.

$$\begin{aligned}
 S_n &= \frac{n}{2} [2a + (n-1)d] \\
 &= \frac{20}{2} [2 \times 13 + (20-1) \times -5] \\
 &= 10 [26 + 19 \times (-5)] \\
 &= 10 [26 - 95] \\
 &= 10 (69) = 690 \\
 &= 690
 \end{aligned}$$

Hence, the value of 690.

Question.3 If $\operatorname{cosec} A = \frac{17}{8}$, then calculate $\tan A$.

Solution.

Simplify the expression,

$$\begin{aligned}
 \operatorname{cosec} A &= \frac{17}{8} \\
 \operatorname{cosec} A &= \frac{H}{B}
 \end{aligned}$$

Using the Pythagoras theorem,

$$H = 17$$

$$B = 8$$

$$p^2 (\text{perpendicular}) = 17^2 - 8^2 = 289 - 64 = 225$$

$$p = 15$$

then,

$$\tan A = \frac{P}{B}$$

$$\tan A = \frac{15}{8}$$

Hence, the value of $\tan A = \frac{15}{8}$.

Question.4 Write the trigonometric ratio of $\sin A$ in terms of $\cot A$.

Solution.

Simplify the expression,

We know that

$$\tan A = \frac{\sin A}{\cos A}$$

Then similarly,

$$\cot A = \frac{\cos A}{\sin A}$$

$$\cos A : \sin A$$

Hence, the ratio is $\cos A : \sin A$.

Question.5. If the ratio of corresponding medians of two similar triangles are 9:16, then find the ratio of their areas.

Solution.

Simplify the expression,

We know that

$$\triangle ABC \sim \triangle DEF$$

$$\angle A = \angle D$$

$$\text{Let } \angle B = \angle E$$

$$\angle C = \angle F$$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{DF}$$

Then,

$$\frac{\text{area}(\triangle ABC)}{\text{area}(\triangle ADF)} = \frac{(AB)^2}{(DE)^2}$$

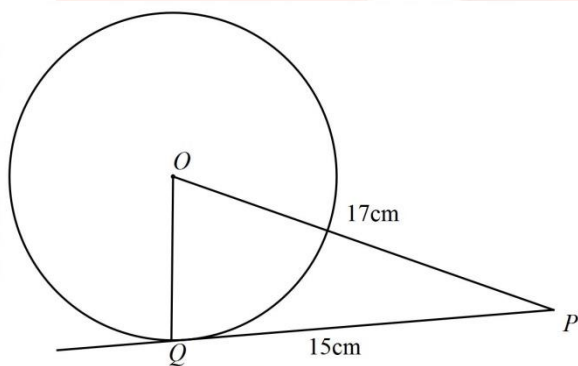
$$\frac{\text{area}(\triangle ABC)}{\text{area}(\triangle ADF)} = \frac{(9)^2}{(16)^2} = \frac{81}{256}$$

$$\frac{\text{area}(\triangle ABC)}{\text{area}(\triangle ADF)} = \frac{81}{256}$$

$$\text{area}(\triangle ABC) : \text{area}(\triangle ADF) :: 81 : 256$$

Question.6 From a point Q, the length of the tangent to a circle is 15 cm and the distance of Q from the center of circle is 17 cm, then find the radius of the circle.

Solution.



Simplify the expression,

We have the length of tangent of 15cm and the distance of Q from the centre of circle is 17cm.

$$QP = 15\text{cm}$$

$$OP = 17\text{cm}$$

then $OQ = ?$

$$OQ^2 = OP^2 - QP^2$$

$$OQ^2 = 17^2 - 15^2$$

$$OQ^2 = 289 - 225$$

$$OQ^2 = 64$$

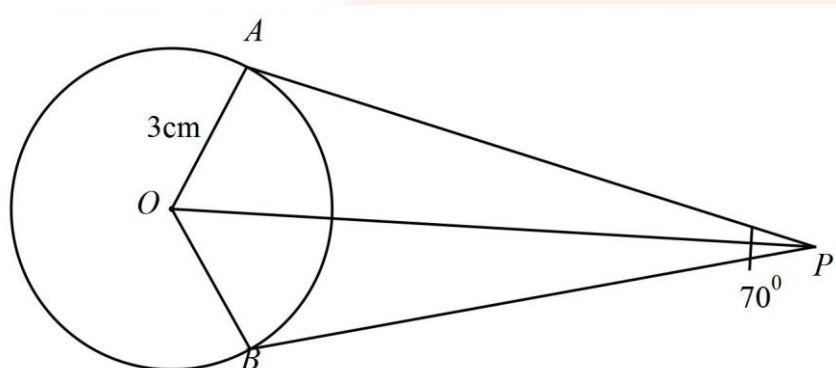
$$OQ = 8\text{cm}$$

Hence, the radius of the circle 8cm.

Question.7 Draw a pair of tangent to a circle of radius 5cm which are inclined to each other at an angle of 70° .

Solution.

Drawn the figure in below.



Hence, the figure drawn above.

Question.8 If the circumference and the area of a circle are numerically equal, then find the radius of the circle.

Solution.

Simplify the expression,

We know that

The circumference of circle is $2\pi r$ and area of circle is πr^2 .

$$2\pi r = \pi r^2$$

$$2 = r$$

$$r = 2$$

Hence, the radius of circle is 2.

Question.9 Find the area of a quadrant of a circle whose circumference is 44cm.

Solution.

Simplify the expression,

We know that the circumference of circle is

$$2\pi r$$

$$2\pi r = 44$$

$$2 \times \frac{22}{7} \cdot r = 44$$

$$r = 7$$

Then,

The area of circle is πr^2

$$\pi r^2 = \frac{22}{7} \times (7)^2$$

$$= \frac{22}{7} \times 7 \times 7$$

$$= 22 \times 7$$

$$= 154m^2$$

Hence, the area of circle is $154m^2$.

Question.10 IF probability of “not E” = 0.95, then find P(E).

Solution.

Simplify probability expression,

$$P(\bar{E}) = 1 - P(E)$$

$$0.95 = 1 - P(E)$$

$$P(E) = 1 - 0.95$$

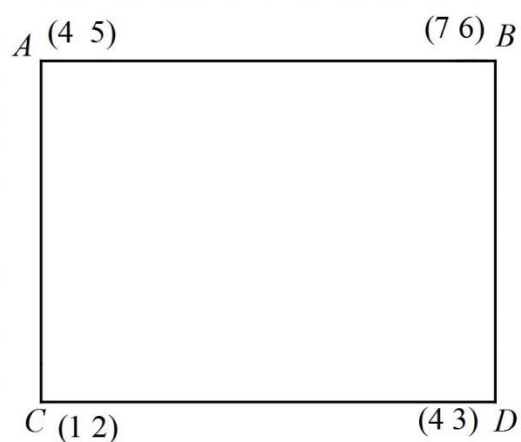
$$P(E) = 0.05$$

Hence, the probability $P(E) = 0.05$.

PART -B

Question.11 Name the type of quadrilateral formed by the points (4, 5), (7, 6), (4, 3), (1, 2).

Solution



Simplify the expression,

Given the points (4, 5), (7, 6), (4, 3), (1, 2).

Formula is distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$AC = \sqrt{(4-7)^2 + (5-2)^2}$$

$$AC = \sqrt{9+9}$$

$$= \sqrt{18}$$

$$= 3\sqrt{2}$$

$$BD = \sqrt{(6-3)^2 + (7-4)^2}$$

$$BD = \sqrt{9+9}$$

$$BD = \sqrt{18} = 3\sqrt{2}$$

$$AB = \sqrt{(6-5)^2 + (7-3)^2}$$

$$AB = \sqrt{1+9}$$

$$AB = \sqrt{10}$$

$$CD = \sqrt{(3-2)^2 + (4-1)^2}$$

$$CD = \sqrt{10}$$

then,

$$AC = BD$$

$$AB = CD$$

Hence, the quadrilateral form is rectangle.

Question.12 A vessel in the form of a hollow hemisphere. The diameter of the hemisphere is 14 cm. find the inner surface area of the vessel.

Solution

Simplify the expression,

We know that the formula is hemisphere of curved surface area is $2\pi r^2$.

dim ater = 14cm

then,

$$\text{radius} = \frac{14}{2} = 7\text{cm}$$

$$2\pi r^2 = 2 \times \frac{22}{7} \times (7)^2$$

$$= 2 \times \frac{22}{7} \times 7 \times 7$$

$$= 2 \times 22 \times 7$$

$$= 44 \times 7$$

$$= 308\text{cm}^2$$

Hence, the inner surface area value is 308cm^2 .

Question.13 Following the mean daily pocket allowance by using appropriate method.

Class interval	10–20	20–30	30–40	40–50	50–60
Frequency s	3	5	4	7	6

Find the mean daily pocket allowance by using appropriate method.

Solution.

And, $\sum f_1 x_1 = f_1 \times x_2$

C.I	f_1	x_1	$f_1 x_1$
10-20	3	15	45
20-30	5	25	125
30-40	4	35	140
40-50	7	45	315
50-60	6	55	330
	$\sum f_1 = 25$		$\sum f_1 x_1 = 955$

$$\text{mean} = \frac{\sum (f_i \times x_i)}{\sum f_i}$$

$$\text{mean} = \frac{955}{25} = 38.2$$

Question.14 A car travels 260 km distance from a place A to place B, at a uniform speed 65km/hr passed through all thirteen green traffic signals, 4 minutes at first signal, 7 minutes at second signal, 10 minutes at third signal and so on stops for 40 minutes at thirteenth signal. How much total time it takes to reach at the place B? Solve by suitable Mathematical Method.

Solution.

Simplify the expression,

We know that

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Then,

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{time}(t) = \frac{\text{distance}}{\text{speed}}$$

$$\text{speed}(s) = 65 \text{ km/h}$$

$$\text{distance}(d) = 260 \text{ km}$$

$$t = \frac{260}{65} = 4 \text{ h}$$

According to the question first signal 4 minutes, second signal 7 minutes, third signal 10 minutes and so stop for 40 minutes at thirteenth signal.

$$1(\text{signal}) = 3$$

$$2(\text{signal}) = 7$$

$$3(\text{signal}) = 13$$

.....

.....

$$13(\text{signal}) = 40$$

$$\text{Difference of time} = 2(\text{signal}) - 1(\text{signal}) = 7 - 4 = 3$$

Then,

$$13(\text{signal}) = 40$$

$$13 \text{ signals stop for total time} = 3 \times 40 = 120 \text{ minutes} \\ = 2h$$

Total time of takes to reach at place B

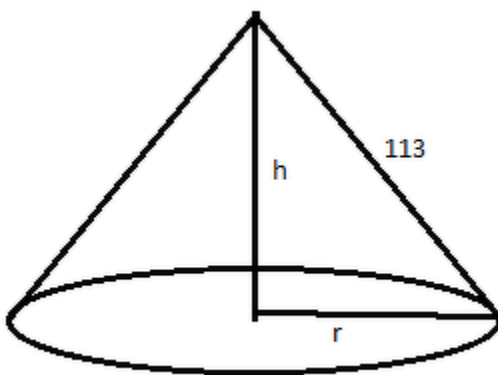
$$T = 4h + 2h = 6h$$

$$T = 6h$$

Hence, the total time takes reach at place B is 6h.

Question.15 For traffic control, a CCTV camera is fixed on a straight and vertical pole. The camera can see 113 m distance straight line from the top. If the area visible by the camera around the pole is 39424m^2 , then find the height of the pole.

Solution.



Here,

$$l = 113$$

$$\pi r l = 39424$$

$$\frac{22}{7} \times r \times 113 = 39424$$

$$r = 111$$

now,

$$l = \sqrt{r^2 + h^2}$$

$$l^2 = r^2 + h^2$$

$$h^2 = l^2 - r^2$$

$$h = \sqrt{113^2 - 111^2}$$

$$h = 21.16m$$

Hence, the value is $h = 21.16m$.

PART-C

Question.16 Prove that $\sqrt{3}$ is an irrational number.

Solution.

Simplify the expression,

Suppose $\sqrt{3}$ is a Rational number.

$$\sqrt{3} = \frac{a}{b} \text{ then, } b \neq 0$$

Now, squaring both side,

$$(\sqrt{3})^2 = \left(\frac{a}{b}\right)^2$$

$$\frac{3}{1} = \frac{a^2}{b^2}$$

$$3b^2 = a^2 \dots\dots\dots(1)$$

$\therefore a^2$, divided by 3

a also divided by 2 then,

$$\frac{a}{2} = \frac{c}{1}$$

$$a = 2c$$

squaring both side,

$$a^2 = 4c^2$$

$$3b^2 = 4c^2$$

$$c^2 = \frac{3b^2}{4}$$

b^2 also divisional by 3/4

so, $\sqrt{3}$ is irrational number.

Question-17 Divided $x^3 - 6x^2 + 11x - 6$ by $x - 2$, and verify the division algorithm.

Solution.

Simplify the expression ,

We know that

$$\text{Dividend} = (\text{Quotient} \times \text{Divisor}) + \text{Remainder}.$$

First the terms of dividend and divisor in decreasing order of their degrees and then perform the division as shown below.

$$\begin{array}{r}
 -x^2 + 4x - 3 \\
 x - 2 \overline{) x^3 - 6x^2 + 11x - 6} \\
 \underline{-x^3 + 2x^2} \\
 -4x^2 + 11x \\
 \underline{+4x^2 - 8x} \\
 +3x - 6 \\
 \underline{-3x + 6} \\
 0
 \end{array}$$

When, remainder is 0.

Hence, the value is $-x^2 + 4x - 3$

Question.18 A manufacturer of TV set produced 720 set in fourth year and 1080 sets in sixth year. Assuming that the production increase uniformly by a fixed number every year, then find total production in first 9 years.

Solution.

Simplify the expression,

720 set in fourth year

1080 set in sixth year

2year difference value of

sixth year -4 year = $1080 - 720 = 360$

2year difference = 360

1 year produced is $= \frac{360}{2} = 180$

Then,

1 year produced is = 180

so,

9 year total production are = $180 \times 9 = 1620$

9 year total production are = $180 \times 9 = 1620$

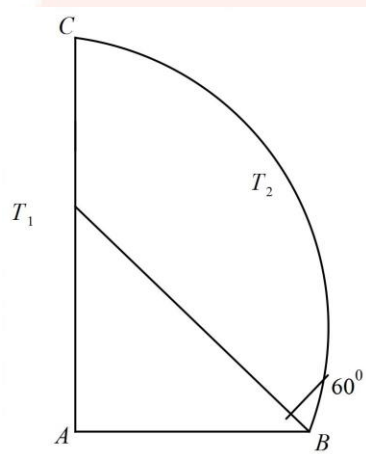
9 year total production are = $180 \times 9 = 1620$

9 year total production are = 1620

Hence, the total production 9 years are 1620.

Question.19 A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 60° with it. The distance between the feet of the tree to the point where the top touches the ground is 3 m. Find the height of the tree.

Solution.



Simplify the expression,

Let T_1 height of before broken tree the tree and T_2 height after broken tree.

$$T_1 + T_2 = ?$$

$$\sin 60^\circ = \frac{T_1}{T_2}$$

$$\frac{T_1}{T_2} = \frac{\sqrt{3}}{2}$$

$$2T_1 = \sqrt{3}T_2$$

$$T_1 = \frac{\sqrt{3}T_2}{2}$$

taking $\cos \theta$

$$\cos 60^\circ = \frac{3}{T_2} = \frac{1}{2}$$

$$T_2 = 6$$

$$T_1 = \frac{\sqrt{3}T_2}{2} = \frac{6\sqrt{3}}{2}$$

$$T_1 = 3\sqrt{3}$$

Hence, the height of tree is $T_1 = 3\sqrt{3}$.

Question.20 Prove the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of the contact the centre.

Solution.

Prove the expression,

Let us consider PT and QT are the tangents line from an external point T. which touches the circle at P and Q respectively.

O is the center of this circle OP and OQ are radius of this circle.

$OP \perp PT$ [angle between radius and tangent at point of contact is 90°]

$$\angle OPT = 90^\circ \dots\dots(i)$$

$OQ \perp TQ$ [angle between radius and tangent at point of contact is 90°]

$$\angle OQT = 90^\circ \dots\dots(ii)$$

Now in quadrilateral POQT

$$\angle OPT + \angle PTQ + \angle OQT + \angle QOP = 360^\circ$$

$$90^\circ + 90^\circ + \angle PTQ + \angle QOP = 360^\circ$$

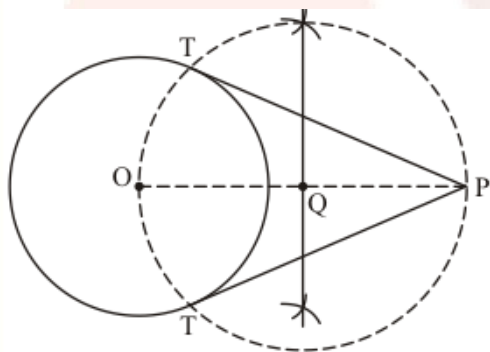
$$\angle PTQ + \angle QOP = 360^\circ - 180^\circ = 180^\circ$$

Since the angle between the two tangent drawn from an external point to a circle is supplementary to the angle subtended by the line-segment.

Joining by the points of contact at the center.

Question.21 Draw a circle of radius 5cm. From a point 13cm away from its centre, construct the pair of tangents to the circle and measure their length. Also verify the measurement by actual calculation.

Solution.



Simplify the expression,

Given that

Construct a circle of radius is 5cm.

Let a point P=13cm

Step: 1 –First of all we draw a circle of radius

OP=13cm

Step: 2- Make a point P at a distance of

OP=13cm

And, join

OP

Step: 3- Taking Q as centre and radius

$$OQ=PQ$$

Draw a circle to intersect the given circle at T and T'

Step: V- joins PT and PT' to obtain the require tangents.

Thus,

PT and P' T'

Are the required tangent Find the length of tangents

We know that

$$OT \perp PT$$

And

$$\Delta OPT$$

$$PT^2 = OP^2 - OT^2$$

$$= 13^2 - 5^2$$

$$= 169 - 25$$

$$= 144$$

$$PT = 12$$

Hence, the length of tangents 12cm

Question.22 In an of a circle subtends an angle of 60° at the centre and if the area of minor sector is 231cm^2 , then find the radius of the circle.

Solution.

Simplify the expression,

We know that

$$\text{Minor sector of area} = \frac{\pi r^2 \theta}{360^\circ}$$

$$= \frac{\pi r^2 \theta}{360^\circ}$$

$$231 = \frac{\pi r^2 60^\circ}{360^\circ}$$

$$231 = \frac{22/7 \times r^2}{6}$$

$$231 = \frac{22 \times r^2}{42}$$

$$22r^2 = 231 \times 42$$

$$r^2 = \frac{231 \times 42}{22}$$

$$r^2 = 21 \times 21$$

$$r = 21\text{cm}$$

Hence, the value of radius is 21cm .

Question.23 A well of diameter 7m and earth from digging is evenly spread out to form a platform $22\text{m} \times 14\text{m} \times 2.5\text{m}$. Find the depth of the well.

Solution.

Simplify the expression,

Volume of platform = volume of cylinder.

$$22 \times 14 \times 2.5 = \pi \left(\frac{7}{2} \right)^2 \times h$$

$$22 \times 14 \times \frac{25}{10} = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times h$$

$$22 \times 14 \times \frac{5}{2} = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times h$$

$$\frac{50}{10} = \frac{1}{4} \times h$$

$$h = 20\text{m}$$

Hence, the height of platform is $h = 20\text{m}$.

Question.24 The following data gives the information on the observed life times (in hours) of 200 electrical components.

Life times (in hours)	40-60	60-80	80-100	100-120	120-140	140-160
Frequency	25	38	65	24	31	17

Determine the modal lifetimes of the components.

Solution.

Simplify the modal expression,

As the life time (in hours) 80-100 has maximum frequency, so it is the modal life time in hours.

$$x_k = 80, h=20, f_k = 65, f_{k-1} = 38 \text{ and } f_{k+1} = 24$$

$$\text{Mode, } M_0 = x_k + \left\{ h \times \frac{(f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})} \right\}$$

$$= 80 + \left\{ 20 \times \frac{(65 - 38)}{(2 \times 65 - 38 - 24)} \right\}$$

$$= 80 + \left\{ 20 \times \frac{27}{(130 - 62)} \right\}$$

$$= 80 + \left\{ 20 \times \frac{27}{68} \right\}$$

$$= 80 + (7.94)$$

$$= 87.94$$

Hence, the value is 87.94.

Question.25 A box contains 7 red marbles, 10 white marbles and 5 green marbles. One marble is taken out of the box at random. What is the probability that the marble taken out will be.

- (i) Not red? (ii) White? (iii) Green?

Solution.

Simplify the probability expression,

Total number of all possible outcomes = total numbers of balls $\Rightarrow 10 + 7 + 5 = 22$

- (i) Total number of not red balls $= 10 + 5 = 15$

Probability (getting a ball which is not red) = $\frac{15}{22}$.

(ii) Number of white balls 10

Probability (getting a white ball) = $\frac{10}{22} = \frac{5}{11}$.

(iii) Number of green ball 5

Probability (getting a green ball) = $\frac{5}{22}$.

Hence, the probability are $\frac{15}{22}$, $\frac{5}{11}$, $\frac{5}{22}$.

Question.26 The cost of 7 erasers and 5 pencils is Rs 58 and the cost of 5 erasers and 6 pencils is Rs 56. Formulate this problem algebraically and Solve the it graphically.

Solution.

Simplify the expression,

Let , cost of 1 erasers =Rs X

Cost of 1 pencils = Rs Y

Then,

$$7x + 5y = \text{Rs } 58 \dots\dots\dots(i)$$

$$5x + 6y = \text{Rs } 56 \dots\dots\dots(ii)$$

Equation (i) multiply with 6 and equation (ii) multiply with 5.

Simplify further,

$$7x + 5y = 58 \dots\dots\dots \times (6)$$

$$5x + 6y = 56 \dots\dots\dots \times (5)$$

$$42x + 30y = 348$$

$$25x + 30y = 280$$

Subtract equation (i) of equation (ii)

$$42x + 30y = 348$$

$$25x + 30y = 280$$

$$17x = 68$$

$$x = 4$$

Putting the value of x in equation (i)

$$7x + 5y = 58$$

$$7 \times 4 + 5y = 58$$

$$28 + 5y = 58$$

$$5y = 58 - 28$$

$$5y = 30$$

$$y = 6$$

$$x = 4$$

$$y = 6$$

Hence, one cost of erasers is 4 and one cost of pencils is 6.

Question.27 A train travels 300 km at a uniform speed. If the speed had been 10km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

Solution.

Let train speed is x .

Then,

$$\text{Time to travel 300 km} = \frac{300}{x}$$

$$\text{Time with speed } (x+10) = \frac{300}{(x+10)}$$

Simplify the further,

$$\frac{300}{x} - \frac{300}{(x+10)} = 1$$

$$\frac{(x+10)300 - 300x}{x(x+10)} = 1$$

$$300x + 3000 - 300x = 1 \cdot x(x+10)$$

$$3000 = x^2 + 10x$$

$$= x^2 + 10x - 3000$$

$$= x^2 + 60x - 50x - 3000$$

$$= x(x+60) - 50(x+60)$$

$$= (x+60)(x-50)$$

$$\Rightarrow x+60 \Rightarrow x = -60 \text{ km/h}$$

$$\Rightarrow (x-50) \Rightarrow x = 50 \text{ km/h}$$

Hence, the speed of train is 50 km/h .

OR

Question. A diagonal of rectangular field is 25 metres more than the shorter side. If longer side is 23 metres more than the shorter side, find the sides of the field.

Solution.

Simplify the expression,

Let, rectangular shorter side is b and longer side is l .

Then,

$$\sqrt{l^2 + b^2} = b + 25 \dots\dots\dots(i)$$

$$l = b + 23 \dots\dots\dots(ii)$$

Equation (i) taking square.

$$[\sqrt{l^2 + b^2}]^2 = (b + 25)^2$$

$$l^2 + b^2 = b^2 + 625 + 50b$$

$$l^2 - 50b - 625 = 0 \dots\dots\dots(iii)$$

Putting the value of l in equation (iii).

$$(b+23)^2 - 50b - 625 = 0$$

$$b^2 + 529 + 46b - 50b - 625 = 0$$

$$b^2 - 4b - 96 = 0$$

$$b^2 - 12b + 8b - 96 = 0$$

$$b(b-12) + 8(b-12) = 0$$

$$(b-12)(b+8)$$

$$b-12 = 0$$

$$\Rightarrow b = 12$$

$$\Rightarrow b+8 = 0$$

$$\Rightarrow b = -8$$

$$l = (b+23)$$

$$l = (12+23)$$

$$l = 35$$

$$b = 12$$

Hence, Rectangular shorter side is 12 and longer side is 35.

Question.28 (i) Evaluate, $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta)$

Solution.

Simplify the expression,

$$(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta)$$

$$\left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right) \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right)$$

$$\left(\frac{\cos \theta + \sin \theta + 1}{\cos \theta}\right) \left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta}\right)$$

$$\frac{(\cos \theta + \sin \theta)^2 - 1^2}{\cos \theta \sin \theta}$$

$$= \frac{\cos^2 \theta + \sin^2 \theta + 2 \cos \theta \sin \theta - 1}{\cos \theta \sin \theta} \quad [\because \sin^2 \theta + \cos^2 \theta = 1]$$

$$= \frac{1 + 2 \cos \theta \sin \theta - 1}{\cos \theta \sin \theta} = \frac{2 \cos \theta \sin \theta}{\cos \theta \sin \theta} = 2$$

$$\Rightarrow 2$$

Hence, the value of 2.

(ii) Prove that, $\frac{\tan A - \sin A}{\tan A + \sin A} = \frac{\sec A - 1}{\sec A + 1}$

Solution.

Prove the expression,

LHS

$$\begin{aligned} & \frac{\tan A - \sin A}{\tan A + \sin A} \\ & \frac{\frac{\sin A}{\cos A} - \sin A}{\frac{\sin A}{\cos A} + \sin A} \\ & = \frac{\sin A - \sin A \cos A}{\sin A + \sin A \cos A} \\ & = \frac{\sin A(1 - \cos A)}{\sin A(1 + \cos A)} = \frac{1 - \frac{1}{\sec A}}{1 + \frac{1}{\sec A}} \\ & = \frac{\sec A - 1}{\sec A + 1} \end{aligned}$$

Hence, proved it.

Question.29 Find the area of that triangle whose vertices are (-3, -2), (5, -2) and (5, 4). Also prove it is right angle triangle.

Solution.

Simplify the expression,

We know that,

$$\text{Slope, } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\begin{aligned}
 &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(5 - (-3))^2 + ((-2) - (-2))^2} \\
 &= \sqrt{(8)^2 + (0)^2} \\
 &= \sqrt{64 + 0} \\
 &= 8
 \end{aligned}$$

$$\begin{aligned}
 |\overline{BC}| &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(5 - 5)^2 + (4 - (-2))^2} \\
 &= \sqrt{0 + (6)^2} = 6
 \end{aligned}$$

$$\text{Area} = \frac{1}{2} |\overline{AB}| |\overline{BC}| = \frac{1}{2} \times 8 \times 6 = 24$$

Since, the value of right angle triangle is (8, 6, and 10).

We get the area value of 24 then, we prove that is right angle triangle.

Question. 30 Prove that a line drawn through the mid-point of one side of a triangle parallel to second side bisects the third side.

Solution.

Prove the expression,

Given:

$\triangle ABC$

$DE \parallel BC$

$AD = DB$ [D is mid-point]

To prove:

$AE = EC$

Proof:

In $\triangle ABC$, $DE \parallel BC$

By the results of B.P.T

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{AD}{AD} = \frac{AE}{EC}$$

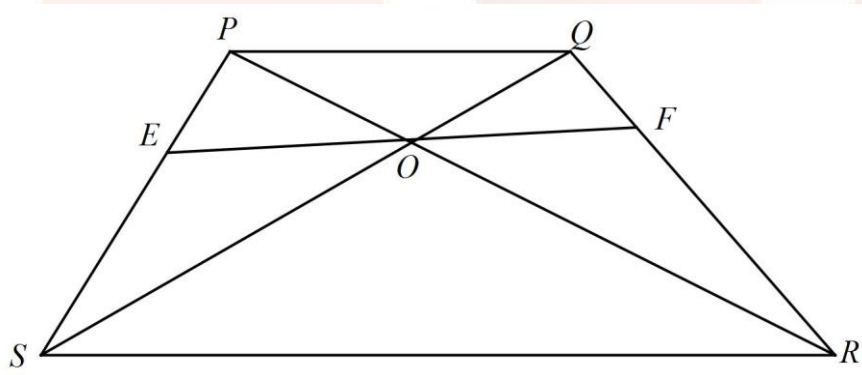
$$AE = EC$$

Hence, prove it.

Question.30 PQRS is a trapezium in which $PQ \parallel RS$ and its diagonals intersect each other at the point O.

Prove that, $\frac{PO}{QO} = \frac{RO}{SO}$.

Solution.



Prove the expression,

To prove: $\frac{PO}{QO} = \frac{RO}{SO}$

Construction: $EOF \parallel PQ \parallel RS$

Proof:

In $\triangle PSR$
 $EO \parallel SR$

By using the result of B.P.T

$$\frac{PE}{ES} = \frac{PO}{RO} \dots\dots(i)$$

In ΔPQS , $EO \parallel PQ$

By using the results B.P.T

$$\frac{PE}{ES} = \frac{QO}{SO} \dots\dots(ii)$$

Equation by (ii).

$$\frac{PO}{QO} = \frac{RO}{SO}$$

by alternendo,

$$\frac{PO}{QO} = \frac{QO}{SO}$$

Hence, prove it.